Supplementary Information

A whole-brain *Drosophila* connectome model predicts a duration- and sign-dependent potassium trap after anoxia and a reset-based route to functional recovery

Srihari Muralikrishnan (Kalaris Labs). Preprint v1.0, 2026-10-20. Contact: hello@kalarislabs.com

All tables below were extracted programmatically from the analysis result files (`research/structure/STRUCTURE_RESULTS.md`, `research/dynamics/DYNAMICS_RESULTS.md`, `research/followup/mechanism/MECHANIS RESULTS.md`, `research/followup/replication/REPLICATION_RESULTS.md`, `research/sim/README.m research/lit/parameters.csv`, `research/followup/replication/tables/v4_condition_summary.csv`, `research/followup/mechanism/raw/sign_res/`, `research/followup/index/tables/` by `build_paper.py`. The claim-by-claim synthesis is `research/SYNTHESIS.md`. They are not retyped. Every value describes the model, not real tissue.

S1. Preregistration (registered 2026-10-16, before any insult simulation)

The full text is in `research/lit/preregistration.md`. Summary of hypotheses and falsification criteria:

H	Key observable	Supported if	Falsified if
H1 sharp threshold	FII(f), Reach(f)	logistic preferred by ΔAIC ≤ −10, 10–90% width ≤ 0.10, irrecoverable beyond f_c	width > 0.25, or ΔAIC > −2, or recovery at f ≥ f_c + 0.10
H2 targeted vulnerability	ρ(demand, death time); f_c,targeted/f_c,random	ρ ≤ −0.3 and ratio ≤ 0.7 (CI excludes 1)	ρ > −0.1 and ratio CI includes 1
H3 graded reperfusion	ΔFII staged − abrupt	≥ +0.05 (CI excludes 0) in ≥2/3 severities, null negative control	ΔFII ≤ 0, or benefit survives without excitotoxic term
H4 hypothermia	Q10_window	within [0.67, 1.5] × model Q10	D_max ratio CI includes 1, or outside band
H5 identity first	D50 ordering; regime coverage	D50_FII lowest by ≥0.05; regime ≥5% of grid	D50_FII not lowest and coverage < 1%
H6 replication	outcome concordance	≥3/4 connectomes match; f_c max/min ≤ 1.5	≤2/4 match, or f_c max/min > 2

Operational "brain death" (preregistered): <1% of neurons spontaneously active for ≥1 s under baseline drive, no readout response (< 2 SD of noise), persisting after the protocol ends. As run (deviation S6): spontaneous fraction < 0.01 and no readout response > 8 Hz after the protocol. This is a model analogue of the clinical construct [Greer et al. 2023], not a clinical determination.

S2. Simulator constants

Table S2a. Fast LIF layer and calibration (from `research/sim/README.md`)

constant	value	source
V_rest = V_reset	−52 mV	Shiu et al. 2024
V_th	−45 mV	Shiu et al. 2024
τ_m / τ_syn	20 ms / 5 ms	Shiu et al. 2024
refractory / delay	2.2 ms / 1.8 ms	Shiu et al. 2024
w_syn	0.275 mV published; **0.185 mV used** (Princeton ≥5 matrix supercritical at 0.275)	calibrated here
Poisson kick	68.75 mV to v	Shiu et al. 2024
dt	0.1 ms (fast), 5 ms (slow layer)	—

readout (mean Hz)	princeton5 w0.275	princeton5 w0.185	shiu783
MN9, sugar 50 / 100 / 200 Hz	65.6 / 162.4 / 195.6	3.3 / 117.7 / 148.3	15.8 / 70.6 / 111.8
DNp01 L/R, LC4 100 Hz	220 / 236	213 / 229	–
DNg100 R, LC9 100 Hz	124	116	–
DNa02 L/R, left LC9+DNa02 180 Hz	197 / 0	n.r.	–

Table S2b. Slow metabolic/ionic layer (phenomenological, not fitted to *Drosophila* data)

constant	default	calibrated set (dynamics, mechanism)	LHS range (mechanism arm L)
τ_ATP	20 s	**30 s**	15–60 s (log)
ATP reserve m	4 × resting demand	4	2–8 (log)
demand exponent β (area^β)	0.5	0.5	0.25–1
r_half (rate demand)	100 Hz	100 Hz	—
pump failure e_ok	0.75	0.75	—
K_0 / K_max	5 / 80 mM	5 / 80 mM	K_max 50–100 mM
g_K (Nernst coupling)	0.6	0.6	0.3–0.9
k_rel (SD release)	60 mM/s	60	20–120 (log)
k_clr (clearance × pump)	1 s^{-1}	1	0.5–2 (log)
V_sd	−30 ± 2 mV	same	—
depolarization block	h ⟶ 1 − σ((v_rest,eff + 32)/2), τ_h 10 ms	same	—
injury hazard	h_max·J^4/(J^4 + $J50^4$), h_max 1/30 s^{-1}	—	—
J50	60 s	**600 s**	300–1,200 s (log)
Q10 demand / supply / pump / injury	2.5 / 1.5 / 1.8 / —	2.5 / 1.5 / 1.8 / **2.5**	not varied
speedup (slow-time compression)	—	**50**	not varied (25–100 tested in arm A)
glutamate sign	inhibitory	inhibitory	inhibitory or excitatory
w_syn	0.185	0.185	0.15–0.25

Table S2c. Calibration runs (dynamics arm A)

condition	SD onset (model s)	dead	FII_norm
tau20_sp25_D60_reset	9.0	0.000	1.001
tau20_sp50_D60_reset	9.5	0.000	1.001
tau20_sp100_D60_reset	10.0	0.000	0.998
tau30_sp25_D60_reset	11.5	0.000	1.001
tau30_sp50_D60_reset	12.5	0.000	1.001
tau30_sp100_D60_reset	13.5	0.000	0.998
tau40_sp25_D60_reset	14.0	0.000	1.001
tau40_sp50_D60_reset	15.5	0.000	1.001
tau40_sp100_D60_reset	16.5	0.000	0.998
J50_60_D120_reset	12.5	0.907	0.131
J50_300_D120_reset	12.5	0.017	0.951
sham_naive	–	0.000	1.001
sham_reset	–	0.000	1.001
sp25_D30_naive	11.5	0.000	-0.104
sp100_D30_naive	13.5	0.000	0.065

S3. Literature parameter table (Table S1; `research/lit/parameters.csv`, 81 rows, 6 flagged UNVERIFIED)

Values are as extracted from the cited sources. "UNVERIFIED" rows were not used as calibration targets.

parameter	value	unit	species	context	source_doi	confidence
K_o_rest_fly_brain	~10	mM	D. melanogaster (+4 Drosophila spp.)	Resting CNS [K+]o, K+-selective microelectrode, cold and warm acclimated	10.1242/jeb.179598	high (full text)
K_o_rest_fly_brain_alt	6.6 ± 0.3	mM	D. melanogaster w1118 males	Baseline [K+]o before repetitive N2 anoxia	10.1242/jeb.074468	high (abstract)
K_o_rest_fly_brain_ouabain_prep	13.1 ± 2.8	mM	D. melanogaster	Baseline between ouabain-induced repetitive surges	10.1371/journal.pone.0098841	high (full text)
K_o_SD_peak_cold_warm_acclimated	58.3 ± 3.7	mM	D. melanogaster (25 °C acclimated)	Peak [K+]o during cold-induced SD	10.1242/jeb.179598	high (full text)

parameter	value	unit	species	context	source_doi	confidence
K_o_SD_peak_cold_cold_acclimated	44.4 ± 5.7	mM	D. melanogaster (15 °C acclimated)	Peak [K+]o during cold-induced SD	10.1242/jeb.179598	high (full text)
K_o_SD_peak_species_range	29.5 – 57.9	mM	Drosophila spp. (montana … birchii)	Peak [K+]o at cold SD; ~3–5x resting	10.1242/jeb.179598	high (full text)
K_o_terminal_plateau	54.1 ± 13.9	mM	D. melanogaster	Terminal loss of K+ homeostasis after repetitive ouabain comas (no recovery; death)	10.1371/journal.pone.0028944	high (full text)
K_hemolymph_fold_change_2h_anoxia	~3	fold	D. melanogaster adult	Hemolymph [K+] after 2 h N2 anoxia; restored on reoxygenation	10.1242/jeb.177147	high (abstract+full text)
t_anoxic_SD_onset_RT	14.2 ± 0.3	s	D. melanogaster (25 °C acclimated)	Time to half-amplitude DC drop after pure N2, room temperature	10.1242/jeb.179598	high (full text)
t_anoxic_SD_onset_RT_cold_acclimated	16.2 ± 0.6	s	D. melanogaster (15 °C acclimated)	As above	10.1242/jeb.179598	high (full text)
t_anoxic_SD_onset_species_range	10.5 – 19.8	s	Drosophila spp.	Time to anoxic SD at room temperature	10.1242/jeb.179598	high (full text)
N2_pulse_threshold_for_K_surge	between 5 and 90 (exact value not extracted)	s	D. melanogaster	N2 pulse duration stepped 5 s⟶90 s revealed a threshold for the [K+]o surge	10.1371/journal.pone.0028944	UNVERIFIED exact value (figure only)
t_paralysis_anoxia_adult	<30	s	D. melanogaster adult	Behavioural paralysis after onset of anoxia	10.1242/jeb.177147	high (abstract)
t_loss_of_coordination_anoxia	60 – 120	s	D. melanogaster	Loss of coordination/falling in N2	10.1016/S0022-1910(96)00084-4	high (abstract)
anoxia_duration_with_full_recovery	up to 240	min	D. melanogaster	Flies tolerated N2 up to 4 h and recovered; nonlinear duration–recovery relation	10.1016/S0022-1910(96)00084-4	high (abstract)
recovery_time_after_<2h_anoxia	<=15	min	D. melanogaster	First movement after reoxygenation (Krishnan 1997 as cited in Campbell 2018)	10.1242/jeb.177147	medium (secondary citation)
recovery_time_after_4h_anoxia	~60	min	D. melanogaster	Mobility recovery time (Krishnan 1997 as cited in Campbell 2018)	10.1242/jeb.177147	medium (secondary citation)
LT50_anoxia_adult	8.3	h	D. melanogaster adult	Anoxia survival, 50% lethal time	10.1242/jeb.125849	high (abstract)
LT50_anoxia_larva	~1	h	D. melanogaster L3	Anoxia survival	10.1242/jeb.125849	high (abstract)
survival_2h_anoxia_adult	100	%	D. melanogaster adult	Despite ATP 3% of normal	10.1242/jeb.177147	high (abstract)
ATP_fraction_15min_anoxia	0.35	fraction of normoxic	D. melanogaster adult	Whole-body ATP (not brain-specific)	10.1242/jeb.177147	high (full text)
ATP_fraction_30min_anoxia	~0.03	fraction of normoxic	D. melanogaster adult	Whole-body ATP; then stable to ~8 h, further drop beyond	10.1242/jeb.177147	high (full text)
tau_ATP_depletion_anoxia	~14	min	D. melanogaster adult	DERIVED: single-exponential fit to 1.0⟶0.35 in 15 min (tau=15/ln(1/0.35)); decline is faster than single exponential by 30 min	10.1242/jeb.177147	derived (assumption-dependent)
metabolic_rate_in_anoxia	~3	% of normoxic	D. melanogaster adult and larva	Metabolic suppression when immobile in anoxia	10.1242/jeb.125849	high (abstract)
O2_consumption_low_O2	~20	% of control	D. melanogaster	O2 consumption in low-O2 atmospheres	10.1016/S0022-1910(96)00084-4	high (abstract)
T_cold_SD_warm_acclimated	5.3 ± 0.5	°C	D. melanogaster (25 °C acclimated)	Temperature of cold-induced SD, 1 °C/min ramp	10.1242/jeb.179598	high (full text)

parameter	value	unit	species	context	source_doi	confidence
T_cold_SD_cold_acclimated	2.4 ± 0.4	°C	D. melanogaster (15 °C acclimated)	As above	10.1242/jeb.179598	high (full text)
T_heat_coma_species_range	38.69 – 45.11	°C	Drosophila spp. (immigrans … mojavensis)	Heat coma (Tcoma) under 0.25 °C/min ramp; coincides with CNS SD	10.1242/jeb.218750	high (full text, bioRxiv version)
T_heat_SD_D_melanogaster	not extracted	°C	D. melanogaster	Species-specific heat-SD temperature reported in figures/tables only	10.1242/jeb.218750	UNVERIFIED
static_heat_SD_assay_temperature	38	°C	Drosophila spp.	Constant-temperature exposure producing SD and heat coma within 1 h	10.1242/jeb.218750	high (full text)
SD_propagation_velocity_RT	2.5 ± 0.3	mm/min	D. melanogaster (warm acclimated)	KCl-induced SD, two-electrode measurement	10.1152/ajpregu.00040.2025	high (full text)
Q10_SD_propagation_velocity	2.98 (warm acclimated); 2.13 (cold acclimated)	dimensionless	D. melanogaster	Temperature sensitivity of SD propagation	10.1152/ajpregu.00040.2025	high (full text)
Q10_SD_onset_slope	~1.6	dimensionless	D. melanogaster	Temperature sensitivity of SD onset slope	10.1152/ajpregu.00040.2025	high (full text)
Q10_metabolic_rate_general	~2.2 – 2.4 (at 15–35 °C)	dimensionless	general (ectotherms)	DERIVED from Boltzmann factor with assumed E=0.65 eV; E value not read from the retrieved abstract	10.1126/science.1061967	UNVERIFIED (derived)
K_o_baseline_drift_repetitive_anoxia_17C	-1.1 ± 1.3	mM (cumulative Δ)	D. melanogaster w1118 males	Total baseline variation during repetitive anoxia at 17 °C	10.1242/jeb.074468	high (abstract)
K_o_baseline_drift_repetitive_anoxia_23C	17.3 ± 1.4	mM (cumulative Δ)	D. melanogaster w1118 males	As above, 23 °C	10.1242/jeb.074468	high (abstract)
K_o_baseline_drift_repetitive_anoxia_29C	332.7 ± 83.0	mM (cumulative Δ)	D. melanogaster w1118 males	As above, 29 °C (cumulative sum, not a concentration)	10.1242/jeb.074468	high (abstract)
energy_share_action_potentials	47	% of signalling energy	rodent grey matter	Attwell–Laughlin budget	10.1097/00004647-200110000-00001	high (abstract)
energy_share_postsynaptic	34	% of signalling energy	rodent grey matter	Attwell–Laughlin budget	10.1097/00004647-200110000-00001	high (abstract)
energy_share_resting_potential	13	% of signalling energy	rodent grey matter	Attwell–Laughlin budget	10.1097/00004647-200110000-00001	high (abstract)
energy_share_cortex_updated	postsyn 50 / AP 21 / rest 20 / release 5 / recycling 4	% of signalling energy	rodent cortex	Updated budget with efficient APs	10.1038/jcbfm.2012.35	high (abstract)
max_fraction_simultaneously_active	<=15	% of neurons	rodent cortex (theory)	Energy-limited distributed coding prediction	10.1097/00004647-200110000-00001	high (abstract)
AP_energy_variation_between_neuron_types	>10	fold	vertebrate and invertebrate models	Depends on Na+/K+ current overlap	10.1371/journal.pcbi.1000840	high (abstract)
t_anoxic_depolarization_mammal	2 – 5	min after severe ischaemia onset	mammals (experimental)	Onset of potentially reversible anoxic depolarisation; injury begins with it	10.1002/ana.25147	high (abstract)
t_terminal_SD_human	3.9 (IQR 2.6–6.3)	min after final perfusion drop	human (dying patients)	Terminal SD after withdrawal of life support	10.1002/ana.25147	high (abstract)
ptiO2_nonspreading_depression_human	11 (IQR 7–14)	mmHg	human	Tissue pO2 at onset of electrical silencing	10.1002/ana.25147	high (abstract)

parameter	value	unit	species	context	source_doi	confidence
CBF_threshold_protein_synthesis	0.55	ml/g/min	mammal	Ischaemic flow threshold hierarchy	10.1002/ana.410360404	high (abstract)
CBF_threshold_anaerobic_glycolysis	0.35	ml/g/min	mammal	As above	10.1002/ana.410360404	high (abstract)
CBF_threshold_energy_failure	~0.20	ml/g/min	mammal	Transmitter release and energy disturbance begin	10.1002/ana.410360404	high (abstract)
CBF_threshold_anoxic_depolarization	<0.15	ml/g/min	mammal	Anoxic depolarisation	10.1002/ana.410360404	high (abstract)
K_o_peak_anoxic_depolarization_mammal	~60 (commonly cited)	mM	mammal	Not read in retrieved text (Hansen 1985 closed access)	10.1152/physrev.1985.65.1.101	UNVERIFIED
window_irreversible_injury_mammal_normothermic	~5 (classical rule)	min	mammal	Classical estimate; not verified in retrieved text and contested by Hossmann 1973 and BrainEx	10.1001/archneur.1973.00490300037004	UNVERIFIED
glycolytic_flux_increase_ischaemia	4 – 7	fold	mouse brain	After decapitation ischaemia	10.1016/S0021-9258(18)51740-3	high (abstract via record)
neuron_loss_rate_stroke	1.9e6	neurons/min	human	Untreated large-vessel ischaemic stroke	10.1161/01.STR.0000196957.55928.ab	high (abstract)
stroke_evolution_duration	10 (6–18)	h	human	Onset to infarct completion	10.1161/01.STR.0000196957.55928.ab	high (abstract)
therapeutic_hypothermia_target	32 – 34 for 24 h	°C	human	After VF cardiac arrest	10.1056/NEJMoa012689	high (abstract)
hypothermia_favourable_outcome	55 vs 39 (RR 1.40, 95% CI 1.08–1.81)	% patients	human	Hypothermia vs normothermia, 6-month neurological outcome	10.1056/NEJMoa012689	high (abstract)
postconditioning_effective_schedule	3 cycles 30 s reperfusion/10 s occlusion; 10 cycles 10 s/10 s	s	rat (focal ischaemia)	Reduced infarct; gradual reperfusion also protective	10.1002/jnr.21703	high (abstract)
postconditioning_onset_deadline	<3	min after reperfusion	rat	Protection lost when started 3 min after reperfusion	10.1002/jnr.21703	high (abstract)
postconditioning_late_window	up to 6	h after focal ischaemia	rodent (review)	Reported delayed postconditioning protection	10.1038/jcbfm.2009.13	medium (review abstract)
BrainEx_postmortem_interval	4	h	pig (isolated brain)	Warm post-mortem interval before ex vivo perfusion	10.1038/s41586-019-1099-1	high (full text)
BrainEx_perfusion_duration	6	h	pig	Normothermic perfusion; cellular/synaptic activity restored, no global ECoG	10.1038/s41586-019-1099-1	high (full text)
OrganEx_warm_ischaemia	1	h	pig (whole body)	Warm ischaemia before OrganEx perfusion	10.1038/s41586-022-05016-1	high (abstract)
LIF_V_rest	-52	mV	D. melanogaster model	Shiu et al. whole-brain LIF	10.1038/s41586-024-07763-9	high (full text)
LIF_V_reset	-52	mV	D. melanogaster model	As above	10.1038/s41586-024-07763-9	high (full text)
LIF_V_threshold	-45	mV	D. melanogaster model	As above	10.1038/s41586-024-07763-9	high (full text)
LIF_R_membrane	10	kOhm cm2	D. melanogaster model	As above	10.1038/s41586-024-07763-9	high (full text)
LIF_C_membrane	2	uF/cm2	D. melanogaster model	As above	10.1038/s41586-024-07763-9	high (full text)
LIF_tau_membrane	20	ms	D. melanogaster model	DERIVED: Cm x Rm = 2 uF/cm2 x 10 kOhm cm2	10.1038/s41586-024-07763-9	high (derived from full-text values)
LIF_tau_synapse	5	ms	D. melanogaster model	Synaptic conductance decay	10.1038/s41586-024-07763-9	high (full text)
LIF_delay	1.8	ms	D. melanogaster model	Spike-to-PSP delay	10.1038/s41586-024-07763-9	high (full text)
LIF_refractory	2.2	ms	D. melanogaster model	As above	10.1038/s41586-024-07763-9	high (full text)

parameter	value	unit	species	context	source_doi	confidence
LIF_W_syn	0.275	mV per synapse	D. melanogaster model	Single free parameter	10.1038/s41586-024-07763-9	high (full text)
LIF_protocol	30 trials x 1000 ms; Poisson input to sensory neurons (e.g. 10–200 Hz sugar GRNs); baseline firing 0 Hz	—	D. melanogaster model	Brian2 implementation	10.1038/s41586-024-07763-9	high (full text)
FAFB_neurons	139255	neurons	D. melanogaster female brain	FlyWire; ~5e7 chemical synapses	10.1038/s41586-024-07558-y	high (abstract)
MCNS_neurons	166700	neurons	D. melanogaster male CNS	11,710 neuron types	10.1016/j.cell.2026.08.015	high (abstract)
NaK_pump_share_total_brain_ATP	~50 (commonly cited); abstract states only that the great majority of energy goes to signalling, particularly Na+ transport	% of brain ATP	mammal	Exact share not stated in the retrieved abstract	10.1016/S0165-0173(00)00038-2	UNVERIFIED (numeric); qualitative claim verified
hypothermia_Bernard_protocol	33 for 12 h	°C	human	Induced within 2 h of ROSC after out-of-hospital cardiac arrest	10.1056/NEJMoa003289	high (abstract)
hypothermia_Bernard_good_outcome	49 vs 26 (adj. OR 5.25, 95% CI 1.47–18.76)	% patients	human	Good outcome at discharge, hypothermia vs normothermia	10.1056/NEJMoa003289	high (abstract)
BD_apnoea_test_targets	pH < 7.30 and PaCO2 >= 60	pH / mmHg	human	No spontaneous respiration at these targets (BD/DNC criterion)	10.1001/jama.2020.11586	high (abstract)
penumbra_survival_time	a few	h	rat (focal ischaemia)	Penumbra disappears into core over a few hours	10.1002/ana.410360404	medium (qualitative, abstract)

S4. Structural percolation (all four connectomes)

Table S4a. Data and pairing QC

dataset	neurons	conn. ids found in annotation	annotated ids present in conn.	pair-level synapse floor in file	pairs (all)	pairs ≥5	sources (sensory)	sinks	of which DN / MN	removable interneurons	modality groups
FAFB	139255	100.00%	99.52%	≥5	3732460	3732460	17550	1415	1305 / 110	120290	22
BANC	158262	100.00%	97.28%	≥3	3037361	1528585	17086	2347	1316 / 1031	138356	50
MCNS	166700	100.00%	99.48%	≥5	6242118	6242118	17937	2241	1320 / 921	146522	41
MANC	23665	100.00%	99.90%	≥1	5305638	1372588	6460	838	0 / 838	16352	21

Table S4b. Baseline reachability (intact, strict)

dataset	Reach_N L4	Reach_N L6	Reach_N ∞	Reach_G L6	Reach_G,loose L6	hops to converge
FAFB	0.4132	0.8687	0.8890	0.9808	0.9866	12
BANC	0.3940	0.7291	0.7869	0.9417	0.9587	18
MCNS	0.6360	0.9019	0.9394	0.9938	0.9960	10
MANC	0.9733	0.9754	0.9754	0.9988	0.9988	10

Table S4c. Damage curves, primary metric Reach_N(L≤6)

dataset	regime	orders	f_c	w (10–90%)	ΔAIC	best alternative	H1 category (prereg)	D50	f_c ratio vs random [95% CI]	D50 ratio vs random [95% CI]
FAFB	random	13	0.800	0.671	-830.1	linear	falsified	0.759	—	—
FAFB	metabolic_area	10	0.277	0.421	-436.2	exponential	falsified	0.288	0.346 [0.337, 0.356]	0.380 [0.371, 0.388]
FAFB	metabolic_syn	5	0.228	0.419	-253.5	exponential	falsified	0.245	0.285 [0.278, 0.291]	0.323 [0.316, 0.330]
FAFB	degree_finegrid	2	0.033	0.231	-35.7	exponential	indeterminate (graded transition)	0.073	0.041 [0.041, 0.041]	0.097 [0.096, 0.098]
FAFB	degree	3	0.035	0.246	-40.2	exponential	indeterminate (graded transition)	0.075	0.044 [0.044, 0.044]	0.099 [0.098, 0.100]
FAFB	degree_matched_random	3	0.035	0.246	-43.8	exponential	indeterminate (graded transition)	0.075	0.044 [0.044, 0.045]	0.099 [0.098, 0.101]
FAFB	focal	5	0.683	1.702	3.8	linear	falsified	0.690	0.854 [0.227, 1.008]	0.910 [0.662, 1.048]
BANC	random	15	0.750	0.734	-829.9	linear	falsified	0.706	—	—
BANC	metabolic_syn	8	0.107	0.232	-269.3	exponential	indeterminate (graded transition)	0.120	0.143 [0.135, 0.150]	0.170 [0.165, 0.176]
BANC	degree_finegrid	2	0.017	0.086	-84.7	exponential	supported (structural criteria)	0.028	0.022 [0.022, 0.023]	0.040 [0.040, 0.041]
BANC	degree	3	0.017	0.083	-147.0	exponential	supported (structural criteria)	0.030	0.023 [0.022, 0.024]	0.042 [0.041, 0.043]
BANC	degree_matched_random	2	0.019	0.081	-120.4	exponential	supported (structural criteria)	0.031	0.025 [0.024, 0.025]	0.043 [0.043, 0.044]
BANC	focal	4	0.031	0.337	2.3	exponential	falsified	0.283	0.041 [0.038, 0.913]	0.401 [0.134, 0.912]
MCNS	random	6	0.951 (extrapolated)	0.744	-184.4	linear	falsified	0.869	—	—
MCNS	metabolic_syn	6	0.304	0.440	-242.4	exponential	falsified	0.316	0.319 [0.309, 0.329]	0.364 [0.358, 0.368]
MCNS	degree_finegrid	2	0.058	0.378	-2.0	exponential	falsified	0.115	0.061 [0.059, 0.062]	0.133 [0.131, 0.134]
MCNS	degree	2	0.076	0.306	-40.1	exponential	falsified	0.115	0.080 [0.077, 0.083]	0.132 [0.131, 0.134]
MCNS	degree_matched_random	2	0.072	0.327	-41.1	exponential	falsified	0.118	0.076 [0.074, 0.078]	0.135 [0.134, 0.136]
MCNS	focal	3	0.028	0.521	2.6	exponential	falsified	0.368	0.029 [0.029, 0.997]	0.423 [0.137, 0.990]
MANC	random	20	1.027 (extrapolated)	0.393	-1045.2	linear	falsified	>0.9	—	—
MANC	metabolic_syn	20	0.677	0.315	-1805.5	linear	falsified	0.673	0.660 [0.644, 0.672]	< 0.748 (random never fell to 50%)
MANC	degree_finegrid	2	0.421 (extrapolated beyond fine-grid max f = 0.3)	0.467	-142.4	linear	falsified	>0.3	0.410 [0.402, 0.418]	n/a (both censored)
MANC	degree	5	0.388	0.412	-572.5	exponential	falsified	0.384	0.377 [0.369, 0.384]	< 0.426 (random never fell to 50%)
MANC	degree_matched_random	5	0.394	0.421	-632.1	linear	falsified	0.396	0.384 [0.375, 0.391]	< 0.440 (random never fell to 50%)
MANC	focal	10	0.847	0.660	-86.9	linear	falsified	0.727	0.825 [0.758, 0.917]	< 0.808 (random never fell to 50%)

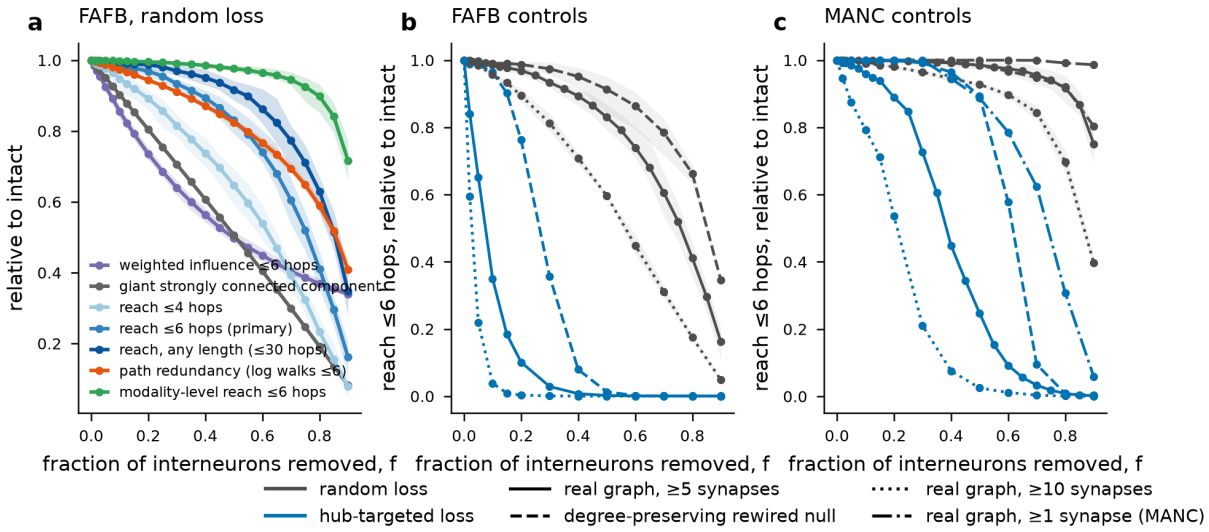


Figure S1. (a) FAFB random loss, all structural metrics; (b, c) degree-preserving null and edge cut-off controls for FAFB and MANC.

Table S4d. D50 (f at 50% of baseline) per structural metric

dataset	regime	weighted influence	GSCC	Reach_N L4	Reach_N L6	Reach_N ∞	redundancy	Reach_G L6
FAFB	random	0.497	0.506	0.626	0.759	0.853	0.858	>0.9
FAFB	metabolic_syn	0.162	0.414	0.133	0.245	0.437	0.302	0.436
FAFB	metabolic_area	0.188	0.433	0.167	0.288	0.468	0.347	0.450
FAFB	degree_finegrid	0.086	>0.3	0.031	0.073	0.266	0.118	0.195
BANC	random	0.295	0.417	0.554	0.706	0.825	0.804	0.862
BANC	metabolic_syn	0.065	0.183	0.071	0.120	0.218	0.173	0.230
BANC	degree_finegrid	0.031	0.108	0.014	0.028	0.065	0.055	0.074
MCNS	random	0.327	0.528	0.806	0.869 (1 cens.)	>0.9	0.896 (1 cens.)	>0.9
MCNS	metabolic_syn	0.091	0.462	0.204	0.316	0.503	0.357	0.490
MCNS	degree_finegrid	0.057	>0.3	0.056	0.115	0.275	0.175	0.237
MANC	random	0.240	0.656	0.884 (1 cens.)	>0.9	>0.9	>0.9	>0.9
MANC	metabolic_syn	0.133	0.584	0.516	0.673	0.759	0.634	0.824
MANC	degree_finegrid	0.091	>0.3	0.223	>0.3	>0.3	>0.3	>0.3

Controls. Degree-preserving rewired null: FAFB random f_c 0.893 (real 0.800), hub-targeted 0.266 (real 0.035); MANC random 1.107 (extrapolated; real 1.027), hub-targeted 0.613 (real 0.388). Edge cut-off ≥10: FAFB random f_c 0.607, w 0.960; hub-targeted f_c 0.013, w 0.087, ΔAIC 0.2 (falsified). MANC ≥1: random f_c 0.787, w 0.124 (indeterminate); hub-targeted f_c 0.770. The H1 category for the hub-targeted regime (FAFB) and the random regime (MANC) is therefore not robust to the edge cut-off. Negative control (off-path neurons removed): Δ = 0 for every metric in every dataset (off-path fractions FAFB 2.1%, BANC 27.0%, MCNS 2.4%, MANC 3.2%).

Keystones. Max relative ΔReach_N(L6) for single-type removal: FAFB 0.88% (CB1231; 154 types evaluated), BANC 1.03% (IN13B010; 218), MCNS 0.86% (AN04A001; 194), MANC 0.00% (161). Top-50 membership overlap did not exceed chance once the universe was restricted to types evaluated in both datasets (p = 0.129–0.634). Graded importance replicated (Spearman ρ of influence loss: FAFB–BANC 0.730, FAFB–MCNS 0.710, BANC–MCNS 0.911, BANC–MANC 0.652, MCNS–MANC 0.776).

S5. Dynamics: brain-death map, protocols, tolerance, neuroprotection

Table S5a. Recoverability boundary (20-s block reset)

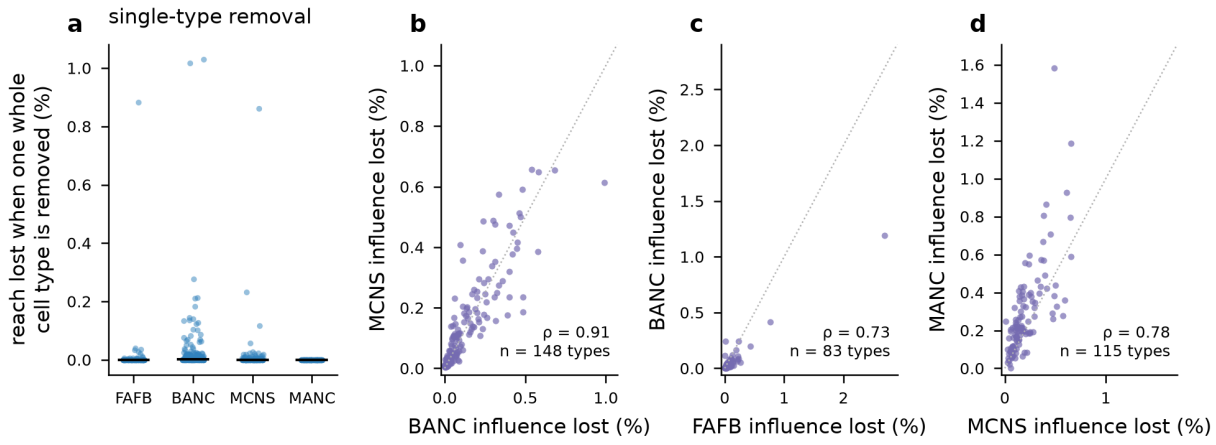


Figure S2. (a) Relative reach lost when all neurons of one cell type are removed (each dot a type; bar = median); (b–d) cross-dataset agreement of relative weighted-influence loss for shared types (Spearman ρ).

T (°C)	SD onset (s)	D_max (model s)	dead at 240 s	outcome at 480 s	brain death from
17	15.0	359 [342, 381]	0.5 %	structural loss (16.8 % dead, FII_norm 0.68)	not reached (960 s: 98.7 % dead, survivors in runaway)
22	12.5	271 [261, 278]	3.8 %	structural loss (66 % dead)	960 s (100 % dead)
29	8.5	149 [145, 153]	40 %	brain death (99.7 % dead)	480 s
33	6.5	124 [112, 129]	81 %	brain death	480 s

Table S5b. All 32 reperfusion protocols, ranked by mean FII_norm (30/180/360 s anoxia, 5 seeds)

rank	protocol	30 s	180 s	360 s	dead @360 s	outcome 30/180/360
1	revive_reset	1.001	1.001	1.001	0.000	recovered/recovered/recovered
2	clr3_reset	1.001	0.963	0.588	0.244	recovered/recovered/structural_loss
3	noise_reset	1.001	0.960	0.585	0.236	recovered/recovered/structural_loss
4	adapt_reset	1.001	0.959	0.577	0.248	recovered/recovered/structural_loss
5	block20_del5	1.001	0.959	0.577	0.248	recovered/recovered/structural_loss
6	staged_fwd	1.001	0.959	0.577	0.248	recovered/recovered/structural_loss
7	block40	1.001	0.959	0.577	0.248	recovered/recovered/structural_loss
8	block20	1.001	0.953	0.578	0.248	recovered/recovered/structural_loss
9	staged_rev	1.001	0.953	0.578	0.248	recovered/recovered/structural_loss
10	sdblock_reset	1.001	0.953	0.577	0.248	recovered/recovered/structural_loss
11	block20_del20	1.001	0.957	0.566	0.250	recovered/recovered/structural_loss
12	block10	1.001	0.959	0.428	0.248	recovered/recovered/runaway
13	adapt	0.734	0.731	0.548	0.248	runaway/runaway/runaway
14	voff-10	0.155	0.361	0.464	0.246	runaway/runaway/runaway
15	voff-20	0.137	0.320	0.435	0.243	runaway/runaway/runaway
16	sdblock	0.409	0.010	0.319	0.252	runaway/runaway/runaway
17	clr3	0.718	-0.170	0.020	0.244	runaway/runaway/runaway
18	ramp10	0.402	0.161	-0.157	0.267	runaway/runaway/runaway
19	gaba4	0.183	0.349	-0.173	0.239	runaway/runaway/runaway
20	noise	0.260	0.015	0.008	0.240	runaway/runaway/runaway
21	naive	-0.058	0.084	0.085	0.252	runaway/runaway/runaway
22	gaba2	0.191	0.100	-0.195	0.239	runaway/runaway/runaway
23	block5	-0.060	0.097	0.018	0.252	runaway/runaway/runaway
24	block2	-0.088	0.098	0.010	0.252	runaway/runaway/runaway
25	naive_noinj	-0.079	-0.078	0.105	0.250	runaway/runaway/runaway
26	pulsed6x5	-0.104	-0.032	-0.123	0.281	runaway/runaway/runaway
27	pulsed3x10	-0.057	-0.093	-0.136	0.273	runaway/runaway/runaway
28	ramp60	-0.104	-0.065	-0.212	0.318	runaway/runaway/runaway
29	ramp30_noinj	-0.305	0.064	-0.175	0.281	runaway/runaway/runaway
30	ramp30	-0.301	0.063	-0.246	0.292	runaway/runaway/runaway
31	cool12	-0.161	-0.127	-0.227	0.260	runaway/runaway/runaway
32	cool17	-0.229	-0.305	-0.161	0.256	runaway/runaway/runaway

Protocol key: `naive` = abrupt step to S = 1; `blockN` = total spike + synaptic block for N s from reperfusion; `block20_delN` = block starting N s after reperfusion; `rampN` = linear supply ramp over N s; `pulsedAxB` = A cycles of B-s reperfusion; `voff-N` = –N mV offset; `gabaN` = N× GABAergic gain; `sdblock` = k_rel × 0.2; `clr3` = 3× clearance; `noise` = low-level background Poisson; `adapt` = spike-frequency adaptation k_adapt 0.2; `coolN` = cooling to N °C during reperfusion; `staged_fwd/rev` = staged release sensory → central → descending (or reverse), 10 s per stage after a 20-s block; `revive_reset` = counterfactual revival of dead neurons + reset; `_noinj` = injury off; `_reset` suffix = followed by 20-s block.

Table S5c. Tolerance and preconditioning (arm D; reset protocol)

condition	SD onset (s)	dead 180 s	FII_norm 180 s	dead 360 s	FII_norm 360 s
hypomet0.2	33.0	0.001	0.995	0.088	0.704 [0.573, 0.784]
hypomet0.5	19.5	0.005	0.983	0.198	0.631 [0.576, 0.662]
hypomet0.8	14.5	0.008	0.970	0.235	0.568 [0.452, 0.657]
precool12	17.0	0.000	1.001	0.006	0.977 [0.966, 0.997]
precool17	15.0	0.001	0.985	0.041	0.908 [0.890, 0.938]
precond15	12.5	0.009	0.961	0.249	0.591 [0.442, 0.681]
arrest0.05	12.5	0.001	0.983	0.059	0.711 [0.608, 0.785]
unprotected (C block20)	12.5	0.009	0.953	0.248	0.578

Table S5d. Targeted (hub) neuroprotection (arm H; 360-s anoxia)

protected set	n neurons	FII_norm (reset)	dead	FII gain per 1,000 protected	FII_norm (naive)
top-degree 1%	1393	0.703 [0.668, 0.743]	0.245	0.090	-0.218
random 1%	1393	0.568 [0.530, 0.591]	0.246	-0.007	0.236
top-degree 2%	2785	0.744 [0.714, 0.791]	0.243	0.059	-0.254
random 2%	2785	0.590 [0.559, 0.617]	0.243	0.004	0.172
top-degree 5%	6963	0.847 [0.789, 0.905]	0.235	0.039	-0.253
random 5%	6963	0.574 [0.531, 0.596]	0.236	-0.001	0.117
top-degree 10%	13926	0.906 [0.845, 0.944]	0.222	0.024	0.342
random 10%	13926	0.588 [0.563, 0.637]	0.224	0.001	-0.118
whole brain	139255	1.001 [1.001, 1.001]	0.000	0.0030	-0.041
none (C block20 / naive)	0	0.578	0.248	–	0.085

S6. Mechanism follow-up

Table S6a. Latin-hypercube ranges (48 sets + calibrated set)

parameter	range
w_syn	0.15–0.25
τ_ATP	15–60 s, log scale
J50	300–1,200 s, log scale
g_K	0.3–0.9
k_rel	20–120, log scale
k_clr	0.5–2, log scale
reserve	2–8, log scale
demand β	0.25–1
K_max	50–100 mM
glutamate	inhibitory or excitatory

Table S6b. Runaway and rescue across parameter sets

anoxia	protocol	sets with runaway (majority of seeds)	sets rescued (FII_norm ≥ 0.8)	mean FII_norm [95 % CI]
30 s	abrupt	**48/48 = 1.00 [1.00, 1.00]**	4/48 = 0.08 [0.02, 0.17]	0.24 [0.14, 0.34]
30 s	abrupt + 20-s block	0/48	**48/48 = 1.00 [1.00, 1.00]**	1.001 [1.000, 1.001]
180 s	abrupt	**48/48 = 1.00 [1.00, 1.00]**	2/48 = 0.04 [0.00, 0.10]	0.17 [0.08, 0.27]

anoxia	protocol	sets with runaway (majority of seeds)	sets rescued (FII_norm ≥ 0.8)	mean FII_norm [95 % CI]
180 s	abrupt + 20-s block	11/48 = 0.23 [0.13, 0.35]	**34/48 = 0.71 [0.58, 0.83]**	0.75 [0.63, 0.85]

Table S6c. Excitotoxic term calibration (180-s anoxia)

k_exc	dead, abrupt	dead, + block
0	0.9 %	0.9 %
0.3	0.9 %	0.9 %
1	1.1 %	0.9 %
3	3.5 %	0.9 %
10	39.5 %	1.0 %

Table S6d. Driver-set connectivity

set (same size)	in largest SCC	excitatory edges per node
sustain drivers	96 %	31
top-degree hubs	40 %	9.4
random	0.4 %	0.1

Table S6e. Targeted 20-s block on held-out seeds

anoxia	sustain drivers	earliest igniters	top-degree hubs	random	whole brain
30 s: smallest rescuing set	**0.5 % (696 neurons), 5/5 seeds, FII_norm 0.947 [0.943, 0.951]**	**0.5 %, 5/5, 0.945**	2 % (2,785), 5/5	none up to 20 % (best 0.33)	1.000
180 s: smallest rescuing set	**20 % (27,851), 5/5, 0.979** (10 %: 0/5)	**20 %, 5/5** (10 %: 0/5)	none up to 20 % (0/5)	none up to 20 % (0/5)	0.979

Table S6f. Dissection (rate in last 10 s of a 40-s manipulation, Hz/neuron, 30 s / 180 s anoxia)

manipulation	rate during manipulation	late rate after release
all synapses off	109 / 134	138 / 120
excitatory synapses off	14 / 120	38 / 91
inhibitory synapses off	133 / 135	111 / 115
K^+ clamped at 5 mM	**2.4 / 2.3**	4.3 / 3.9
no spike-driven K^+ release (k_spk = 0)	**2.4 / 2.3**	4.3 / 4.1
no SD K^+ release (k_rel = 0)	56 / 153	129 / 25

Table S6g. Basin boundary: minimal K^+ kick reaching the high-rate state vs tonic offset δ

offset δ	+6 mV	+4 mV	+2 mV	0 mV	−2 mV	−4 mV	−6 mV
minimal kick	6 mM	7 mM	8 mM	9 mM	12 mM	15 mM	none up to 30 mM
analytic K_th	5.3	6.1	6.9	7.9	9.0	10.2	11.7 mM

S6h. Transmitter-sign robustness (antennal-lobe local neurons inhibitory)

Variant A sets every neuron with class ALLN or primary type matching lLN/vLN/v2LN/ilLN/il2LN to inhibitory. In FAFB the mask covers 431 neurons; 284 were already inhibitory and 147 changed sign (90 predicted cholinergic, 57 predicted monoaminergic). The same rule changed 147 neurons in the FAFB-proxy model and 219 in MCNS.

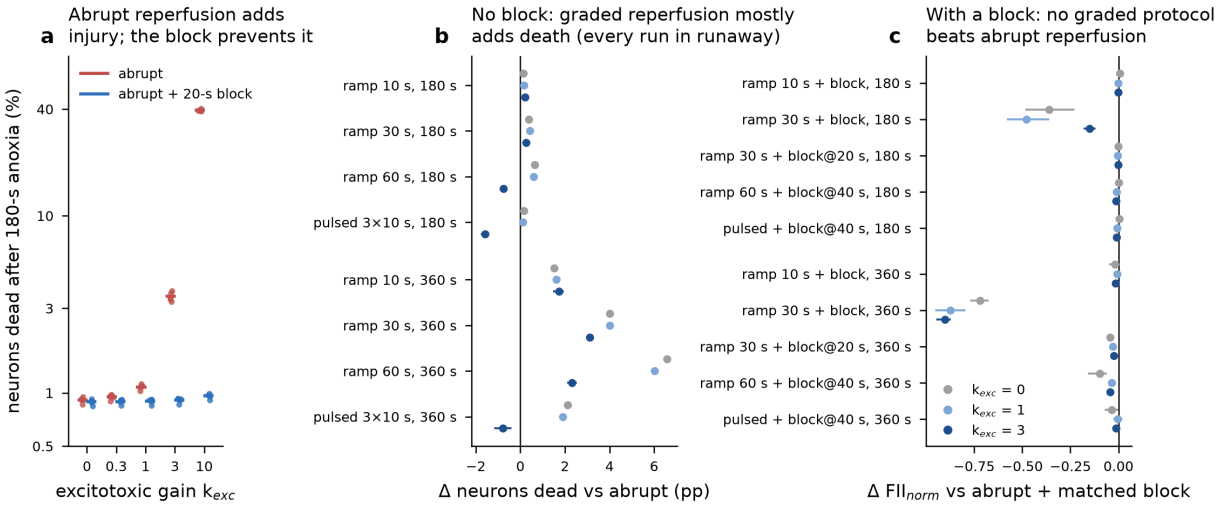


Figure S3. (a) Dead fraction after 180-s anoxia vs excitotoxic gain k_exc (n = 5 seeds per point). (b) Paired Δ dead, graded − abrupt, no block. (c) Paired Δ FII_norm, graded + block − abrupt + matched block. Dots = means with paired-bootstrap 95% CIs.

Table S6h-1. FAFB full model, variant A, 30-s anoxia, naive reperfusion (per seed; local runs). FII is raw (vs the default-sign intact baseline; no variant-A baseline was run); FII_norm ≈ FII/0.9940. Runaway rule: late rate > 0.5 Hz/neuron or FSA > 0.05. Default-sign comparison (dynamics arm C, 5 seeds): FII_norm −0.058, runaway 5/5; anatomy runs N_anat_D30 s0–s4: mean observation rate 91.3–104.3 Hz/neuron, 64.0–69.5 % of neurons ≥ 20 Hz.

run	seed	dead	late rate (Hz/neuron)	FSA	FII (raw)	FII/0.9940	surge peak (Hz)	K^+ peak (mM)	runaway
S_A_D30_naive_s0	0	0.000	0.168	0.0050	0.915	0.920	68.4	44.9	no
S_A_D30_naive_s1	1	0.000	0.160	0.0049	0.928	0.933	69.6	45.0	no
S_A_D30_naive_s2	2	0.000	0.035	0.0008	0.936	0.942	68.9	45.0	no

anatomy run	seed	mean observation rate (Hz/neuron)	fraction of neurons ≥ 20 Hz	median first spike (ms)
SN_A_anat_D30_s0	0	0.172	0.0028	105
SN_A_anat_D30_s1	1	0.160	0.0027	105
SN_A_anat_D30_s2	2	0.035	0.0005	105

Table S6h-2. MCNS and FAFB-proxy, default signs vs antennal-lobe local neurons inhibitory (_`alln`), 5 seeds per condition.

model	condition	n	outcomes	FII_norm [95% CI]	late rate (Hz/neuron)	FSA	dead
FAFBproxy	D180_T22_naive	5	{"runaway": 5}	0.288 [0.086, 0.505]	124.09	0.044	0.009
FAFBproxy_alln	D180_T22_naive	5	{"runaway": 5}	-0.125 [-0.155, -0.093]	126.62	0.743	0.009
MCNS	D180_T22_naive	5	{"runaway": 5}	0.856 [0.810, 0.901]	127.79	0.001	0.009
MCNS_alln	D180_T22_naive	5	{"runaway": 5}	-0.135 [-0.234, -0.041]	120.53	0.897	0.009
FAFBproxy	D180_T22_reset	5	{"recovered": 5}	0.971 [0.961, 0.984]	0.00	0.000	0.009
FAFBproxy_alln	D180_T22_reset	5	{"recovered": 5}	0.965 [0.948, 0.978]	0.00	0.000	0.009
MCNS	D180_T22_reset	5	{"recovered": 5}	0.997 [0.992, 1.001]	0.00	0.000	0.009
MCNS_alln	D180_T22_reset	5	{"recovered": 5}	0.989 [0.970, 1.000]	0.00	0.000	0.009
FAFBproxy	D30_T22_naive	5	{"runaway": 5}	0.146 [0.057, 0.228]	47.11	0.140	0.000
FAFBproxy_alln	D30_T22_naive	5	{"runaway": 5}	0.859 [0.852, 0.867]	0.82	0.008	0.000
MCNS	D30_T22_naive	5	{"runaway": 5}	0.050 [0.022, 0.078]	90.86	0.084	0.000
MCNS_alln	D30_T22_naive	5	{"runaway": 5}	1.000 [0.996, 1.002]	3.00	0.022	0.000
FAFBproxy	D30_T22_reset	5	{"recovered": 5}	1.002 [1.001, 1.003]	0.00	0.000	0.000
FAFBproxy_alln	D30_T22_reset	5	{"recovered": 5}	1.002 [1.001, 1.003]	0.00	0.000	0.000
MCNS	D30_T22_reset	5	{"recovered": 5}	0.999 [0.994, 1.001]	0.00	0.000	0.000

model	condition	n	outcomes	FII_norm [95% CI]	late rate (Hz/neuron)	FSA	dead
MCNS_alln	D30_T22_reset	5	{“recovered”: 5}	1.000 [0.997, 1.002]	0.00	0.000	0.000
FAFBproxy	base	5	{“recovered”: 5}	1.003 [1.001, 1.003]		0.000	0.000
FAFBproxy_alln	base	5	{“recovered”: 5}	1.003 [1.001, 1.003]		0.000	0.000
MCNS	base	10	{“recovered”: 10}	0.999 [0.996, 1.001]		0.000	0.000
MCNS_alln	base	5	{“recovered”: 5}	1.000 [0.997, 1.002]		0.000	0.000

contrast (AL local neurons inhibitory – default signs; closed-loop – 20-s block)	Δ FII_norm [95% CI]
MCNS_alln_minus_main_D30_naive	+0.949 [+0.918, +0.977]
FAFBproxy_alln_minus_main_D30_naive	+0.714 [+0.632, +0.801]
MCNS_alln_minus_main_D180_naive	-0.991 [-1.119, -0.883]
FAFBproxy_alln_minus_main_D180_naive	-0.413 [-0.622, -0.192]
MCNS_closedloop_minus_reset20_D180	-0.071 [-0.081, -0.058]
FAFBproxy_closedloop_minus_reset20_D180	-0.674 [-0.742, -0.624]

Variants B (all low-confidence calls inhibitory) and C (neurons without a call silenced) were prepared in code but not run. FAFB variant A was not run at 180 s, with the reset, for bistability kicks, or for targeted blocks.

S7. Cross-connectome dynamic replication (MCNS v1.0; BANC v888 not analysed)

Source: `research/followup/replication/REPLICATION_RESULTS.md` (211 analysed runs: MCNS 146, FAFB-proxy 65). MCNS: 166,700 neurons, 6,242,085 edges (≥5 synapses), w_syn 0.0538 mV (same distance below the reverberation threshold as FAFB). Proxy substitutions: demand = total synapse count; K^+ coupling = connection-based Jaccard 9-nearest-neighbour neuropil graph. A dense shared-neuron K^+ adjacency was numerically unstable (spurious oscillation and runaway in sham runs) and was replaced; the FAFB dynamics track used the stable geometric adjacency and is unaffected. Responsive probes only (sugar → MN9 is silent in MCNS); noise ceilings MCNS 0.9972, FAFB-proxy 0.9922.

Table S7a. Condition summary

condition	**MCNS** FII_norm [95 % CI] · dead · outcomes	FAFB-proxy	FAFB (dynamics track)
30 s anoxia, naive	0.050 [0.022, 0.078] · dead 0.000 · runaway 5/5	0.146 [0.057, 0.228] · dead 0.000 · runaway 5/5	-0.058 [-0.180, 0.153] · runaway 5/5
30 s, 20-s block	0.999 [0.994, 1.001] · dead 0.000 · recovered 5/5	1.002 [1.001, 1.003] · dead 0.000 · recovered 5/5	1.001 [1.000, 1.001] · recovered 5/5
180 s, naive	0.856 [0.810, 0.901] · dead 0.009 · runaway 5/5	0.288 [0.086, 0.505] · dead 0.009 · runaway 5/5	0.084 [-0.118, 0.410] · runaway 5/5
180 s, 20-s block	0.997 [0.992, 1.001] · dead 0.009 · recovered 5/5	0.971 [0.961, 0.984] · dead 0.009 · recovered 5/5	0.953 [0.936, 0.967] · recovered 5/5
180 s @17 °C, naive	0.915 [0.874, 0.943] · dead 0.001 · runaway 5/5	–	–
180 s @17 °C, block	0.992 [0.984, 0.998] · dead 0.001 · recovered 5/5	–	–
180 s @29 °C, naive	0.483 [0.374, 0.592] · dead 0.128 · runaway 5/5	–	–
180 s @29 °C, block	0.806 [0.738, 0.868] · dead 0.127 · recovered 3/5, structural_loss 2/5	–	–
360 s, naive	-0.033 [-0.275, 0.255] · dead 0.251 · runaway 5/5	–	0.085 [-0.120, 0.311] · runaway 5/5
360 s, block	0.759 [0.691, 0.833] · dead 0.246 · structural_loss 5/5	0.613 [0.562, 0.671] · dead 0.246 · structural_loss 5/5	0.578 [0.559, 0.602] · structural_loss 5/5
360 s hub 1 %, naive	0.235 [0.042, 0.396] · dead 0.249 · runaway 5/5	–	-0.218 [-0.286, -0.093] · runaway 3/3
360 s hub 1 %, block	0.960 [0.948, 0.973] · dead 0.244 · structural_loss 5/5	0.707 [0.678, 0.737] · dead 0.244 · structural_loss 5/5	0.703 [0.668, 0.743] · structural_loss 3/3
360 s hub 10 %, naive	-0.137 [-0.197, -0.071] · dead 0.223 · runaway 5/5	–	0.342 [0.300, 0.378] · runaway 3/3
360 s hub 10 %, block	0.977 [0.972, 0.980] · dead 0.219 · recovered 5/5	–	0.906 [0.845, 0.944] · recovered 3/3
360 s random 1 %, block	0.769 [0.704, 0.837] · dead 0.244 · structural_loss 5/5	–	0.568 [0.530, 0.591] · structural_loss 3/3

condition	**MCNS** FII_norm [95 % CI] · dead · outcomes	FAFB-proxy	FAFB (dynamics track)
360 s random 10 %, block	0.813 [0.775, 0.851] · dead 0.222 · structural_loss 5/5	–	0.588 [0.563, 0.637] · structural_loss 3/3
360 s whole-brain, block	0.999 [0.994, 1.001] · dead 0.000 · recovered 5/5	–	1.001 [1.001, 1.001] · recovered 3/3
180 s, closed-loop release	0.926 [0.919, 0.939] · dead 0.007 · runaway 5/5	0.296 [0.228, 0.346] · dead 0.007 · runaway 5/5	–
sham (0 s), naive	1.001 [1.000, 1.001] · dead 0.000 · recovered 3/3	–	–
sham (0 s), block	1.001 [1.000, 1.001] · dead 0.000 · recovered 3/3	–	–

Table S7b. Hub-protection efficiency in MCNS (360-s anoxia + 20-s block)

protected set (MCNS, n neurons)	gain in FII_norm vs none	gain per 1,000 protected
top-degree 1 % (1,667)	+0.201 [+0.127, +0.278]	0.1208
top-degree 10 % (16,670)	+0.218 [+0.143, +0.293]	0.0131
random 1 %	+0.010 [-0.089, +0.114]	0.0061
random 10 %	+0.054 [-0.028, +0.140]	0.0032
whole CNS (166,700)	+0.239 [+0.165, +0.315]	0.0014

Table S7c. Replication verdicts

test	verdict (MCNS; BANC not analysed)
(a) runaway attractor after naive reperfusion	**replicates** (5/5 at 30, 180, 360 s and 17/22/29 °C; continuous at 30 s, episodic at ≥180 s)
(b) 20-s block reset rescue	**replicates** (5/5 recovered at 30 and 180 s)
(c) hub-protection efficiency	**replicates, stronger** (~84× per neuron vs FAFB ~30×; random controls null)
(d) temperature effect	**replicates in direction** (17 °C − 29 °C = +0.185); magnitude not comparable to FAFB at matched D
VNC vs brain	the VNC crosses 1 Hz/neuron first and is the most persistent part of the attractor

BANC. A BANC sweep (146 result files on the Modal Volume `flysim-data:/results_rep_v4/BANC*`) completed but was not retrieved or analysed because further cloud work was stopped (job recorded as failed; cost unknown, ≤ USD 1.76 by the container cap). MANC was not simulated dynamically.

S8. Recoverability index (cross-validated; FAFB)

Source: `research/followup/index/tables/model_results.json` and `univariate_*.csv`; 1,118 of 1,209 dynamics runs processed (91 not processed); condition-grouped 5-fold cross-validation (4–20 repeats depending on the model) with cluster-bootstrap 95% CIs. Indices were computed on default-sign graphs. Intact FAFB: R_eff (mean-field) 19.62; confident-transmitter R_eff 7.38; signed 17.05 (`index/data/fafb_intact_index.json`). `ecore100` = fraction of neurons in the excitatory bootstrap core (maximal set in which every member receives enough excitatory synapses from other members to sustain firing at an assumed 100 Hz); `recip_n` = surviving reciprocal excitatory pairs relative to intact; `reach_sim` = simulator Reach.

Table S8a. Post-reset FII_norm, non-runaway runs (top single indices)

index	n	CV R^2 [95% CI]	Spearman ρ
ecore100	562	0.964 [0.940, 0.978]	0.949
ecore200	562	0.964 [0.937, 0.980]	0.940
recip_n	562	0.961 [0.934, 0.975]	0.943
reach_sim	562	0.938 [0.891, 0.965]	0.857
hub5	562	0.923 [0.856, 0.955]	0.824

index	n	CV R^2 [95% CI]	Spearman ρ
hub10	562	0.922 [0.857, 0.957]	0.843
hub1	562	0.914 [0.851, 0.950]	0.795
gscc_exc	562	0.908 [0.807, 0.951]	0.920

Table S8b. Runaway after naive reperfusion among viable networks (top single indices)

index	n (runaway)	a-priori AUC [95% CI]	CV AUC	Spearman ρ
exc_in_frac	482 (457)	0.532 [0.450, 0.653]	0.529	0.025
rich1_n	482 (457)	0.526 [0.484, 0.570]	0.525	0.021
EI_syn	482 (457)	0.510 [0.466, 0.549]	0.488	0.008
ipr_exc	482 (457)	0.479 [0.310, 0.790]	0.297	0.016
loop3_super	482 (457)	0.364 [0.174, 0.738]	0.297	-0.109
loop2_super	482 (457)	0.379 [0.194, 0.742]	0.263	-0.098
reach_sim	482 (457)	0.351 [0.158, 0.800]	0.254	-0.118
Reff_mf_known	482 (457)	0.342 [0.146, 0.726]	0.254	-0.124

Combined models: post-reset FII random forest CV R^2 0.986 [0.974, 0.992]; recovered-after-reset logistic CV AUC 0.999; runaway among viable networks random forest CV AUC 0.61 [0.38, 0.76] (chance). Leave-arm-out transfer of `ecore100` for post-reset FII: arm D R^2 0.89, arm H 0.84, arm E (ageing) −0.50. The planned BANC/MCNS/MANC recoverability maps were started (FAFB random-removal grid to f = 0.6 only) and not completed; no standalone INDEX_RESULTS report was written.

S9. Deviations from the preregistration and task plan (all tracks)

1. **Seeds and stimulus rates.** 3–5 seeds per condition (preregistered ≥10); one rate per probe (preregistered 3). Baseline used 10 seeds.
2. **Spontaneous-activity metric.** The Shiu model is silent at rest, so FSA_baseline = 0 and the preregistered ratio is undefined. H5 used an evoked-neuron count (Act).
3. **Brain-death definition as run.** Spontaneous fraction < 0.01 and no readout response > 8 Hz (preregistered: < 2 SD of noise).
4. **Calibration choices made after inspecting runs.** J50 60 ⟶ 600 s, τ_ATP 20 ⟶ 30 s, q10_injury 2.5 added, w_syn 0.275 ⟶ 0.185. All are reported with the rejected alternatives.
5. **H1 dynamical test** was fitted on FII vs dead fraction from insult runs, not on removal curves.
6. **H2(a)** (death-time correlation with demand) was not computed.
7. **H3 protocols** were not matched for integrated supply.
8. **Temperatures** 20 and 26 °C were not run (17/22/29/33 °C were).
9. **Reach in dynamics** used the ≥5 matrix only, with no cut-off sensitivity. Ageing synapse loss is not reflected in Reach.
10. **LT50** was extrapolated, not simulated (≤1 h anoxia simulated).
11. **Structural removal orders** were below the preregistered ≥20 for several regimes (FAFB random 13, metabolic_area 10, metabolic_syn 5, degree 3 + 2 fine-grid, focal 5, matched 3; BANC 15/8/3/4/2; MCNS 6/6/2/3/2; MANC full) because of CPU contention. The f grid was reduced from 24 to 13 points mid-run, L∞ was capped at 30 hops for later orders, and GSCC was computed for orders 0–2 only. The primary metric samples 128 source neurons.
12. **Structural controls reduced.** One rewired null (FAFB, MANC only); ≥1 cut-off only for MANC (FAFB/MCNS files are floored at 5).
13. **A CSV column-shift bug** (rows without GSCC) affected 143 FAFB, 78 BANC and 65 MCNS rows. These were repaired deterministically.
14. **MAOL v1.1** was not available locally and was not analysed (planned deviation).
15. **Mechanism follow-up of 605 runs not executed** (Modal spend limit): 40/60-s blocks for all LHS sets, the full k_exc = 10 H3 grid, finer targeted-set brackets (0.2–0.3% at 30 s; 12.5–17.5% at 180 s), and a hysteresis ramp to −30 mV.

16. **Per-set FII baselines in arm L** used 4 seeds. Arm X k_exc = 0 cells reused dynamics arm C runs (identical code path).
17. **Dynamic replication.** Earlier MCNS sweeps (v1, adjacency bug; v2, infrastructure failure) were discarded. The v4 sweep gave 211 analysed runs (MCNS + FAFB-proxy). The BANC v4 sweep (146 files) was not harvested or analysed. MCNS and FAFB-proxy use proxies for demand and K^+ coupling. MANC was not simulated dynamically.
18. **Recoverability index.** 91 of 1,209 runs were not processed. Cross-connectome maps were not completed. Indices use default signs. No standalone report was written; results are taken from `model_results.json`.
19. **Sign robustness.** Only the antennal-lobe local-neuron variant was run: FAFB full model 3 seeds at 30 s naive only (no variant-A intact baseline, so FII is relative to the default-sign baseline); MCNS and FAFB-proxy 5 seeds at 30/180 s naive and reset. Variants B and C were not run.
20. **Closed-loop release** was tested only in MCNS and the FAFB-proxy (180 s, default signs), not in the FAFB full model or across LHS sets.

S10. Negative and null results (reported in full)

1. No sharp percolation threshold under random structural loss in any of four connectomes (H1 falsified; FAFB w 0.671).
2. No sharp threshold for FII vs dead fraction (ΔAIC +0.11 vs linear; width 1.56).
3. Graded, ramped or pulsed reperfusion was worse than abrupt reperfusion for identity and survival (H3 reversed). Adding an excitotoxic term did not change this for any comparison that included a reset.
4. Hyperpolarization (−10/−20 mV), GABA gain ×2–4, SD-release block, 3× clearance, noise, adaptation and cooling during reperfusion all failed to prevent the runaway.
5. Blocks of 2 and 5 s failed, and 10 s was unreliable at 360 s.
6. Preconditioning gave no protection (24.9% vs 24.8% dead).
7. Hyperthermia without hypoxia (33/37/40 °C) had no effect.
8. Metabolic support (reserve boost, clearance boost, mild hypometabolism) did not rescue the aged network.
9. "Resuscitated but different" did not exist outside the runaway state (0.0% coverage with the reset).
10. No single cell type is a structural keystone (≤1.03% reach loss).
11. Fly-like metabolic arrest alone did not produce hours-scale tolerance (extrapolated LT50 ≈ 1.51 h at J50 14,400 s vs fly 8.3 h).
12. No connectome index predicted naive-reperfusion runaway among viable networks (cross-validated best single-index AUC 0.53; combined 0.61 [0.38, 0.76]).
13. Targeted driver or hub blocks failed after 180 s of anoxia up to 10% of the brain.
14. A closed-loop release at K^+ < 6 mM and ATP > 0.9 (reached after 11 s) re-ignited the runaway in MCNS (runaway 5/5) and the FAFB-proxy.
15. With antennal-lobe local neurons inhibitory, the 30-s high-rate runaway did not form (FAFB 0/3 runaway; MCNS and FAFB-proxy kept a weak 3.0 / 0.82 Hz/neuron oscillation classed runaway by the preregistered rule). The 30-s driver-core result is therefore not established.

S11. Provenance and compute

- Dynamics: Modal jobs a4ddd6e8-9ca3-4803-a889-fa5fdd58ccf9, 31c05682-cf05-4a51-b769-97b562f60172, 4d88396b-d4af-4b98-935c-ea62f206acfb, 937a47ef-8836-46d7-8764-a2a4d3655763 (smoke 79482286-adfb-4c8d-b57c-64d67726799e); 1,209 runs, 0 errors, 24.3 CPU-h.
- Mechanism: smoke 900ff527-e669-4ffa-bcd0-5eed11e9297e; shard A 65780d67-c068-43a9-8cf0-1976a09783a9; shard B c6fe6ce5-53f5-4e4e-afe0-dc930dc25632; 2,061 runs, 0 errors. Follow-up 12a56f59-2ec3-4e3c-8934-253f564c9204 was killed by the spend limit and its results discarded.

- Replication: calibration 6fd134b9-a8a3-4e02-8438-143947ce30ea; v1 MCNS 6e86553b-044e-4155-becd-21845a5eb524 (invalid); v2 04f516ec-6e0a-4889-86d3-d5d9a903a4f6 (infrastructure failure).
- Replication (v4, analysed): smoke 14de3ae1-d0bf-412c-8841-c1a279754447 (MCNS 360-s hub arm, 45 runs); job A a9ef9894-0c76-4de2-89c9-6879f59167ff (166 runs). BANC job ef7d8554-1bbc-4a61-a027-11bb646177dc recorded as failed; 146 files on the Volume, not analysed.
- Sign variant (FAFB full model): local runs, `research/followup/mechanism/raw/sign_res/` (S_A_D30_naive_s0–2, SN_A_anat_D30_s0–2).
- Recoverability index: local; `research/followup/index/`.
- Structure: local 16-core host; pure NumPy bitset engine.
- Identity check: the mechanism-track simulator with k_exc = 0 reproduced dynamics run `C_D30_naive_s0` exactly (FII 0.3617, late rate 122.998 Hz/neuron).
- Run totals in the paper: 1,209 dynamics + 2,061 mechanism (784 of them in the LHS arm) + 211 replication + 6 FAFB sign-variant = 3,487 analysed simulated condition–seed runs.